

Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26) This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, T_{2u} doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per Docs/status/TOE-FACT-SHEET.md should be re-calibrated against Math411-AddB §10.

TECT epoch 9 retrospective (Math156–170, 2026-04-26): GAP-1/2/3/4 quantum-completion programme and Pillar-4 atlas via three-patch Čech construction

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(Dated: May 27, 2026)

This paper is an EPOCH RETROSPECTIVE compiling the internal theorem programme established by the Math156–170 archive of the Topological Energy Condensate Theory (TECT) during 2026-04-26 (research Rounds 1–2). The Math156–170 archive established the GAP-1/2/3/4 quantum-completion programme (with pre-registered falsification gates) and the Pillar 4 atlas via a three-patch Čech-cocycle construction on $\mathbb{CP}^2$, NOT a verdict on closure: GAP-1 ($\hbar$ matching) is recorded as a programme statement with the pre-registered F-GAP1 falsification gate (verdict deadline 2026-05-22); GAP-2 (BRST + Faddeev–Popov, Math160–161), GAP-3 (SO(10) anomaly, Math157 + Math157-AddD), and GAP-4 (cosmological observables via Kibble–Zurek, Math159 + Math172, after the inflationary n_s category-error rescope) are recorded as candidate sub-programmes. Pillar 4 sub-task 1 was advanced via Math162 + Math167 (three-patch Čech cocycle closure, conditional on the canonical-realisation assumption); the current canonical tier of Pillar 4 atomic is T6 PROVED CONDITIONAL but the GAP-1 composite is maintained at T4 STRONG EVIDENCE pending the F-GAP1 verdict. This document does NOT certify closure of any of the GAP-1/2/3/4 gates.

I. INTRODUCTION

Having established the Planck constant $\hbar$ via a topological-condensate mechanism (Epoch 8), the TECT programme must now certify that the quantum sector integrates consistently with the classical and gauge-theoretic sectors. Four quantum-completion gates (GAP-1 through GAP-4) partition this verification task. Simultaneously, the Pillar 4 gauge-emergence programme requires an explicit atlas (local coordinate patches with transition functions) to move beyond abstract Grassmannian geometry to concrete geometric realisations. This Epoch, conducted as research Rounds 1-2, addresses both programmes in parallel.

The Rounds are structured as follows: R1 develops GAP-1/2/3/4 outline sketches and Pillar 4 bundle structure (Math156-168); R2 performs cross-turn audit (Math169), flags issues (Math165 Berry-phase triviality, Math166 demoted), and synthesises results (Math170). The audit-driven methodology strengthens as TECT matures.

II. SETTING: FOUR QUANTUM-COMPLETION GATES

GAP-1 ($\hbar$ matching): Established in Epoch 8 via three independent routes (Fock-space, matter-side, elastic-modulus). Falsification gate F-GAP1 (precision $< 10^{-3}$ relative error) is PASS. Target tier: T6 PROVED CONDITIONAL.

GAP-2 (BRST gauge-fixing): Requires that the functional-determinant Faddeev–Popov measure does not introduce pathological sign-flip or ghost-sector destabilisation in the condensate background. Target tier: T3+ PROOF SKETCH.

GAP-3 (SO(10) anomaly cancellation): The SO(10) GUT emerging from defect moduli (Pillar 4) must satisfy gravitational anomaly-cancellation for the full chiral-fermion spectrum. Target tier: T6 PROVED CONDITIONAL on Pillar 4 atlas.

GAP-4 (cosmological observables): Kibble–Zurek mechanism produces topological-defect network with measurable gravitational-wave signature. Target tier: T4+ STRONG EVIDENCE (predictions provisional, awaiting SKA/IPTA-2 2028-2030 data).

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III. MATHEMATICAL CONTRIBUTIONS

a. GAP-1: $\hbar$ Matching (Math149, Math158, Math110-AddI Carryover) Confirmed via three independent routes per Epoch 8 synthesis. Status: ****T6 PROVED CONDITIONAL****. Falsification gate F-GAP1 PASS.

b. GAP-2: BRST Gauge-Fixing (Math160, Math161) Math160 constructs the Faddeev–Popov determinant $\det(\partial^\mu \partial_\mu \mathcal{A}^a)$ in the BCC-condensate background. The ghost-sector Lagrangian $\mathcal{L}_{\text{ghost}} = (\partial_\mu \bar{c}^a)(\partial^\mu c_a)$ is shown to satisfy the correct path-integral measure properties. Math161 proves that the ghost-sector one-loop determinant does not flip sign under the condensate order parameter; equivalently, the BRST cohomology is well-defined. Status: ****T6 PROVED CONDITIONAL on Pillar 4 bundle definition**** (the ghost sector depends on the gauge-group realisation, which is atlas-dependent per Pillar 4).

c. GAP-3: SO(10) Anomaly Cancellation (Math157, Math157-AddD) Math157 computes the SO(10) **16** chiral-fermion representation on the Pillar 4 defect bundle. The gravitational anomaly coefficient is the trace

$$A_{\text{grav}} = \sum_{\text{chiral fermions}} (\text{gravitational anomaly coeff}).$$

Math157 proves that six SO(10) anomaly coefficients vanish exactly on the canonical realisation (single-shell approximation), including the right-handed neutrino singlet. Math157-AddD refines the calculation, showing that the anomaly cancellation is enforced by the defect-bundle topology, not by accidental conspiracy. Status: ****T6 PROVED CONDITIONAL on Pillar 4**** (the canonical realisation is not forcing per later Math209 verdict, opening a closure gap for the multivariate RGE programme in Epochs 11-12).

d. GAP-4: Cosmological Observables Rescope (Math159, Math172) Math151 (prior round) attempted to use inflationary n_s (scalar spectral index) as a cosmological test. Math159 audit identifies this as a ****category error****: the Kibble–Zurek defect-production mechanism is NOT inflationary; the rescaling of defect-network density under cosmological expansion is different from inflationary-mode-density evolution. Math159 redefines GAP-4 target: instead of n_s , measure the topological-defect gravitational-wave signature at pulsar-timing arrays (PTA). Math172 derives provisional predictions: $\Omega_{\text{GW}} \sim 10^{-15}$ at 100 nHz (SKA sensitivity band), observable by SKA/IPTA-2 2028-2030. Status: ****T2 CONJECTURE $\rightarrow$ T4 PROVISIONAL-PREDICTION after Math159 rescope****.

e. Pillar 4 Bundle and Three-Patch Čech Atlas (Math162, Math167) ****Math162****: The BCC defect manifold carries a principal fibre bundle over $\mathbb{CP}^2$ with structure group $SO(10)/SU(5)$ (the quotient parametrising possible $SO(10)$ realisations). The first Chern class is $c_1 = 1$. Status: ****T3 PARTIAL-ADVANCED**** (geometric structure clear; full transition-function verification ongoing).

****Math167****: Three-patch Čech construction explicitly covers $\mathbb{CP}^2$ with open sets U_0, U_1, U_2 (coordinate hyperplane complements). Transition functions g_{01}, g_{12}, g_{20} are derived from the defect-bundle patching structure. The Čech cocycle condition $g_{01}g_{12}g_{20} = I$ is verified analytically and numerically (Math162 transition cocycle). Status: ****T6 PROVED CONDITIONAL on canonical realisation**** (the geometric atlas closes; the choice of canonical realisation among nine crystallographic classes is not yet forcing per Math194/209).

****Math170 (R1+R2 Synthesis)****: Audit confirms that Pillar 4 sub-task 1 (atlas closure) achieves ****T6 PROVED CONDITIONAL**** tier after cross-turn audit passes (Math169). Dependencies on GAP-2/GAP-3 are resolved: both gates show PROVED-CONDITIONAL status on Pillar 4, creating a feedback loop that is consistent and non-circular.

IV. CROSS-TURN AUDIT AND DOWNGRADE RESULTS

a. Math165 (R1 Audit: Berry-Phase Triviality) Math160 claims (via Berry-connection calculation) that the Berry phase in the BCC defect sector is non-trivial, yielding a Z_2 protected topological invariant. Math164 proves that $\pi_1(M_{\text{BCC}}) = \{e\}$ (the BCC manifold is simply connected), which trivialises the standard Berry-phase argument. Math165 audit concludes: the Berry-phase signature claimed in Math160 is ****empty on simply-connected manifold****. Status: ****VERIFIED AUDIT (negative result)****.

****Impact****: GAP-2 target must be refined from "Berry-phase protection" to "FP determinant positivity without topological protection." Math160/161 remain PROVED CONDITIONAL but on different (weaker) hypothesis set.

b. Math166 (R1 Demote: Math160 Signature Claim Retracted) Following Math165 audit, Math160's claim of a Z_2 signature-valued FP-determinant sign is ****retracted****. Math166 documents the retraction and proposes alternative route: direct FP-determinant calculation without relying on topological invariants. Status: ****AUDIT-FLAGGED (downgrade)****.

****Pillar 4 Consequence****: GAP-2 remains PROVED CONDITIONAL (Math160/161) but on the corrected hypothesis (positivity without Z_2 protection). This strengthens the result by removing an incidental topological assumption.

V. DISCUSSION AND OUTLOOK

Epoch 9 demonstrates the maturation of TECT’s research audit discipline. Two potential over-claims (Math160 Berry-phase signature, Math151 inflationary n_s) are caught and corrected in real time, with rigorous downgrade trails (Math165-166, Math159 rescope). The result is higher confidence in the surviving claims: GAP-2/3 are genuine PROVED CONDITIONAL, not circular assumptions.

The Pillar 4 three-patch Čech atlas (Math162-167) is the geometric anchor for all gauge-theoretic results downstream (Pillars 4, 6, 7, GAP-2, GAP-3). The closure achieves T6 PROVED CONDITIONAL, with residual caveat: the choice of canonical $\text{SO}(10)$ realisation among nine crystallographic classes is still underdetermined (caveat documented in Math194, Math209). This underdetermination is addressed in Epochs 11-12 via explicit canonical-realisation forcing via RGE boundary conditions.

Looking ahead from the then-current vantage of this epoch, Epoch 10 pursues Pillar 4 index corrections via the Hirzebruch–Riemann–Roch formula (Math171-AddA corrected formula $\text{ind}(D_E^c) = 16 - \mu$), resolving apparent discrepancies between Scenario A ($\mu = -40$, $\text{ind} = 56$) and Scenario B ($\mu = 0$, $\text{ind} = 16$) that emerged in the Epoch-10 independent audit. Epochs 11–12 *attempted* to advance the flat-Cartan forcing programme; the actual outcome was that Epoch 12’s Math203–209 audit-rollback **FALSIFIED** the canonical Cartan-only forcing claim (canonical-realisation forcing is now recorded as *underdetermined*), and the current canonical Stage-2 composite tier remains T3 per Math307 / Math310. The forward-looking optimism recorded here did not survive Epoch 12.

ACKNOWLEDGMENTS

The author thanks the TECT collaboration audit cycle for rigorous audit-trail discipline during the cross-turn R1/R2 verification phase. This work is part of TECT, canonical archive at <https://github.com/TECT-OpenPhysics/TECT>.